Cisco Packet Tracer Project 2

Title

Design and Implementation of a VLAN-Based Small Office Network using Router-on-a-Stick (ROAS)

Objective

The objective of this project is to design and implement a small office network for three departments using Cisco Packet Tracer. The network uses Virtual LANs (VLANs) and Router-on-a-Stick (ROAS) to provide communication between departments while maintaining logical network separation.

Project Requirements

1. Create three departments:
    - HR
    - Sales
    - IT
2. Each department should have at least two PCs.
3. Place one network printer in the HR department.
4. Place one server in the IT department.
5. Use:
    - One Router
    - Two Switches
6. Create VLANs for each department.
7. Configure trunk links between networking devices.
8. Configure Router-on-a-Stick for inter-VLAN communication.
9. All VLANs should be able to communicate with each other.
10. All PCs should be able to communicate with the server and printer according to project requirements.

Network Topology

The network consists of:

- One Cisco Router
- Two Cisco Switches
- Two HR PCs
- Two Sales PCs
- Two IT PCs
- One Network Printer
- One Server

The router provides inter-VLAN communication using Router-on-a-Stick technology.

VLAN Configuration

Department	VLAN ID
HR	10
Sales	20
IT	30

End devices are assigned to their respective VLANs using switch access ports.

IP Addressing Scheme

Network: 192.168.50.0/24

Subnetting was performed to provide separate networks for each VLAN.

Example allocation:

HR VLAN:
192.168.50.0/29

Sales VLAN:
192.168.50.8/29

IT VLAN:
192.168.50.16/29

Each VLAN has its own default gateway configured on the router sub-interface.

Devices Used

- 1 Cisco Router
- 2 Cisco Switches
- 6 PCs
- 1 Server
- 1 Network Printer

Network Technologies Used

- IPv4 Addressing
- Subnetting
- VLANs
- Access Ports
- Trunk Ports
- Router-on-a-Stick (ROAS)
- Inter-VLAN Routing
- ICMP Connectivity Testing

Cable Connections

Switch to PC:
Copper Straight-Through Cable

Switch to Server:
Copper Straight-Through Cable

Switch to Printer:
Copper Straight-Through Cable

Router to Switch:
Trunk Connection

Switch to Switch:
Trunk Connection

Configuration Process

1. Design the network topology.
2. Connect all devices using appropriate cables.
3. Create VLAN 10, VLAN 20 and VLAN 30.
4. Assign switch access ports to their respective VLANs.
5. Configure trunk ports between switches.
6. Configure the router interface for Router-on-a-Stick.
7. Create router sub-interfaces for each VLAN.
8. Configure IEEE 802.1Q encapsulation.
9. Assign gateway IP addresses to router sub-interfaces.
10. Configure IP addresses for all end devices.
11. Configure default gateways.
12. Verify connectivity between VLANs.

Connectivity Testing

The following tests were performed:

- HR PC to HR Printer.
- HR PC to Sales PC.
- Sales PC to IT PC.
- IT PC to Server.
- HR PC to Server.
- Sales PC to Server.
- Inter-VLAN communication between all departments.

Successful ping responses confirmed proper VLAN and Router-on-a-Stick operation.

Result

A VLAN-based small office network was successfully designed and implemented using Cisco Packet Tracer. Router-on-a-Stick was configured to provide communication between VLANs while maintaining logical separation of departmental networks.

Conclusion

This project demonstrates practical implementation of VLANs and inter-VLAN routing using Router-on-a-Stick. The project provides hands-on experience with VLAN configuration, trunking, sub-interfaces, IEEE 802.1Q encapsulation, subnetting, and network connectivity testing. The successful completion of this project improves practical networking skills and provides a foundation for more advanced Cisco networking concepts.

Skills Learned

- IPv4 Addressing
- Subnetting
- VLAN Configuration
- Access Port Configuration
- Trunk Port Configuration
- Router-on-a-Stick (ROAS)
- IEEE 802.1Q Encapsulation
- Inter-VLAN Routing
- Default Gateway Configuration
- Network Connectivity Testing using Ping
- Cisco Packet Tracer Network Design